

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(f)		Attempt at $\frac{1}{2}mv^2$ i.e. $0.5\text{ m } v_{\text{max}}^2 = 0.5 (0.030) (0.1)^2(1)$ $= 1.5 \times 10^{-4} \text{ J}(1)$ Extension at equilibrium $(= \frac{mg}{k} = \frac{(0.030)(9.81)}{1.46} = 0.20 \text{ m } (1)$ Adding 1.5 cm [= 0.215 m] (1) Attempt at $\frac{1}{2}kx^2$ i.e. $= 0.5 k x^2 0.5 (1.46)(0.215)^2[1 \text{ ecf}]$ $= 0.034 \text{ [J]}]$ so therefore claim is correct (1)			6	6	6	6
			Question 4 total	6	11	6	23	14	20